

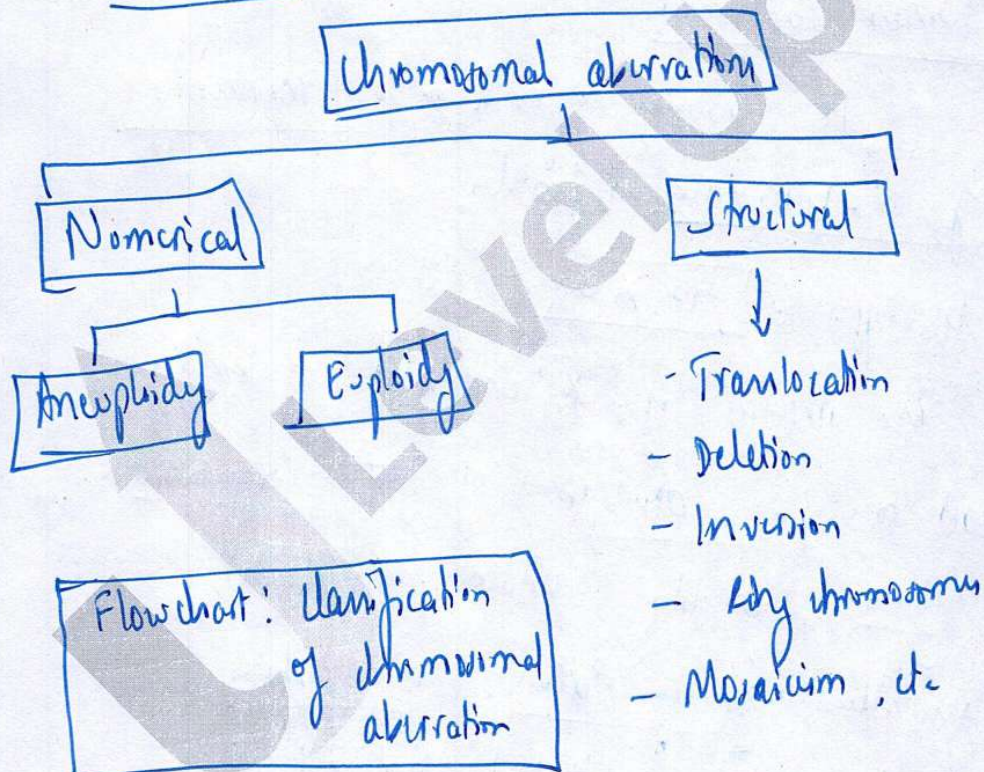
Q.8b (15 marks): Chromosomal aberrations can play havoc with human body and mind. Explain with suitable examples



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- 8.b) Chromosomal aberrations can play havoc with human body and mind. Explain with suitable examples. (15 marks)

Chromosomal aberration refers to changes in structure or number of chromosomes present in human leading to chromosomal disorders.



Play havoc

Genes are basis of inheritance & functional

unit of DNA. Chromosomes made up of genes are crucial for inheritance of several important characters, and their absence or aberration cause disorders, that affect both physical & mental development.

Consider example of

Turner syndrome

It is also referred to monosomy of X chromosome in females

Genotype : 45 X 0

The absence of ^{one} 'X' chromosome in females lead to poor development of female reproductive organs, like undeveloped breasts, genitalia, short stature, microcephaly, & regressive intellectual development, etc.

Downs syndrome

It is referred to as trisomy of 21st chromosome

Genotype: $47 XX | 47 XY$

The down's syndrome children suffer from microcephaly / poor development of brain, epicanthic folds / mongoloid features, autism, poor cognitive abilities, etc.

Klinefelter's syndrome

It is referred to as trisomy of X chromosome in male.

Genotype: ~~47 XY~~ $47 XXY$

They have features like gynecomastia (male breast system), poor male reproductive organ development, longer thighs, poor mental cognition, etc.

Therefore, there is need for genetic counselling & screening as a to be made integral input of modern medicine affordable to all.